



7 Segment Counter Display

Muhammad Saad 202445 | Sohaib Tausif 2024595

Professor Salman Ashraf

CE221: Digital Logic Design

3rd Semester, Batch 34

Department of Cyber Security, GIKI

December 17, 2025

7 Segment Counter Display

Muhammad Saad 2024445 Sohaib Tausif, 2024595

Abstract—This project is focused on the design and implementation of a two-digit 7-segment counter display system which converts internally generated binary-coded decimal (BCD) values into a human-readable numerical output. It uses push-button control to increment the count, which is processed through counter then decoder ICs and finally displayed as a decimal number ranging from 00 to 99 on dual 7-segment displays. The main idea of this project is to demonstrate the practical application of combinational and sequential logic concepts through a reliable hardware-based counting system.

The components used in this project are, two 74190 BCD up/down counter ICs, two 7447 BCD to 7-segment decoder ICs, two common-anode 7-segment displays, a push button, and resistors. A +5V power supply is provided to operate all logic ICs and display units.

The design approach combines both sequential and combinational logic. The 74190 ICs generate BCD outputs that corresponds to the units and tens digits. The 7447 decoder ICs apply Boolean logic internally to activate the appropriate segments of each 7-segment display. This structured design eliminates complex gate-level implementations while maintaining the accurate counting and decoding.

The whole system successfully counts and displays decimal values from 00 to 99, confirming correct operation and design reliability. As an enhancement, the implementation of a two-digit display using cascaded counters, provides a more practical and realistic counting range.

I. INTRODUCTION

The 7-segment displays are one of the most commonly used output devices in digital electronic systems. They are widely found in applications like calculators, counters, digital clocks, and measuring instruments. It consists of seven light-emitting segments arranged in a special pattern that allows the formation of decimal digits from 0 to 9 by selectively switching individual segments ON or OFF. As they have clear visual output and straightforward operation, 7-segment displays are frequently used in educational and practical digital logics to demonstrate number representation and display techniques. In multi-digit systems, multiple 7-segment displays are cascaded to represent even larger numerical ranges.

In digital systems, information is generally processed in binary form, that is not directly understandable to human user and this creates the need for decoding mechanisms that can translate binary or BCD values into a readable decimal format. In counter-based systems, the output of counter ICs is in BCD form, and hence, must be decoded before being displayed. Without decoding, controlling each segment may require complex gate-level logic. Therefore, decoder ICs are used. It simplifies this process by automatically converting BCD outputs into appropriate segment control signals.

7447 BCD to 7-segment decoder IC is used in this project because it is specifically designed to interface with common-anode 7-segment displays. It receives a 4-bit BCD input from

the counter IC and (internally) generates the required segment control signals using combinational logic. This IC reduces hardware complexity, and allows easy integration with TTL-based counter circuits such as the 74190.

The motivation behind choosing this project is to strengthen practical understanding of both combinational and sequential logic systems. By implementing a two-digit automatic counter using hardware components, this project bridges the gap between theoretical concepts of counters, decoders, and displays, and their practical application in the digital systems.

II. OBJECTIVES

A. *Apply combinational logic:*

Understand and apply combinational logic concepts by decoding BCD outputs from counter ICs into corresponding decimal digits using the 7-segment decoder ICs.

B. *Implement hardware using TTL ICs:*

To design and assemble a working digital counter circuit using standard TTL components such as the 74190 counter and 7447 decoder ICs.

C. *Display decimal numbers correctly:*

To accurately display decimal numbers from 00 to 99 on dual 7-segment displays through proper cascading of counter and decoder stages.

D. *Verify results practically:*

To verify the system operation by comparing observed counting and display behavior with expected theoretical results.

III. PROBLEM STATEMENT AND SCOPE

In the digital electronic systems, data is generally processed and represented in binary form. But binary numbers are not easily understandable for human users, especially when multiple bits are involved. For example, the output of a counter IC is a binary or BCD value that does not directly convey any meaningful information. This creates the communication gap between digital systems and users, and so, making it necessary to convert binary data into a clear and readable numerical form.

To address this issue, a reliable method is required. It can translate binary counter outputs into decimal digits that can be displayed visually. BCD provides a great solution by representing each decimal digit using a 4-bit binary number. In an automatic counting system, counter ICs generate BCD outputs that must be decoded before being applied to a display. Without proper decoding giving a 7-segment display from BCD signals is not practical. Therefore, a BCD to 7-segment decoder is used. It automatically generate the correct segment activation pattern for each decimal digit.

The aim of this project is to implementing a two-digit decimal counter capable of displaying values from 00 till 99. The system uses two 74190 BCD counter ICs configured in a cascaded manner to generate units and tens digits. Push

button is used here to provide clock pulses for counting, allowing the display to increment automatically. The decoded outputs are shown on two common-anode 7-segment displays using 7447 decoder ICs. Unlike simple combinational systems, this project incorporates sequential logic through counters, and still maintaining a clear and structured design suitable for educational purposes.

IV. REAL-LIFE APPLICATIONS

A. Digital counters:

Two-digit 7-segment counter systems are widely used in digital counters i.e event counters, frequency counters, and laboratory instruments (where numerical values need to be displayed clearly).

B. Clocks:

Digital clocks depends on cascaded counters and 7-segment displays to represent time values.

C. Control panels:

Industrial and electronic control panels use multi-digit 7-segment displays to show the status codes, operating parameters, and numerical settings for monitoring & control purposes.

V. THEORY

A. Binary Coded Decimal:

Binary Coded Decimal is a digital representation method in which each decimal digit is encoded separately using a 4-bit binary number. Unlike pure binary systems, where values are represented as a continuous binary count, BCD preserves the decimal structure, that makes it easier to interface with display devices. For instance, the decimal digit 7 is represented as 0111 in BCD. Only combinations from 0000 to 1001 are considered valid in BCD, corresponding to decimal digits 0 through 9. This representation is widely used in digital counters because it simplifies the decoding process and it also allows direct conversion to human-readable decimal form without complex binary-to-decimal logic.

B. 7-Segment Display (Segments a–g):

A 7-segment display consists of seven light-emitting segments labelled a, b, c, d, e, f, g arranged in a rectangular pattern. By selectively glowing these segments, decimal digits from 0 to 9 can be formed. Each digit requires a unique combination of activated segments. Like, displaying the digit “0” requires all segments except g to be turned ON. Due to this structured arrangement, 7-segment displays are easy to handle using decoder circuits. Their clear visibility, low power consumption, simple interfacing make them suitable for digital counters, clocks, and, measurement instruments.

C. Common Anode vs Common Cathode Displays:

7-segment displays are available in two main types: common anode & common cathode. In a common anode display, all the anode terminals of the LED segments are connected and tied to the positive supply

voltage. Individual segments are illuminated by providing a LOW signal to their cathode terminals. On the other hand, a common cathode display connects all cathodes to ground, and segments are turned ON by giving a HIGH signal. This project uses common-anode 7-segment displays as they are directly compatible with the 7447 decoder IC, (which provides active-LOW outputs suitable for driving such displays).

D. Common Anode vs Common Cathode Displays:

7-segment displays are available in two main types: common anode & common cathode. In a common anode display, all the anode terminals of the LED segments are connected and tied to the positive supply voltage. Individual segments are illuminated by providing a LOW signal to their cathode terminals. On the other hand, a common cathode display connects all cathodes to ground, and segments are turned ON by giving a HIGH signal. This project uses common-anode 7-segment displays as they are directly compatible with the 7447 decoder IC, (which provides active-LOW outputs suitable for driving such displays).

E. 7447 IC Working:

The 7447 is a BCD to 7-segment decoder. It converts a 4-bit BCD input into seven output signals that control the individual LED segments of the display.

F. Inputs (A, B, C, D):

The IC accepts four input lines labelled A, B, C, and D, which represent a single BCD digit. These inputs are provided by the output pins of the 74190 counter IC.

G. Active-LOW Outputs:

The outputs of the 7447 are active-Low, means a logic LOW signal turns a display segment ON, and a logic HIGH turns it OFF. This matches the operating requirement of common-anode displays.

H. Internal Decoding Logic:

Inside the IC, combinational logic circuits decode the BCD input and generate the correct pattern of LOW signals. The decoder eliminates the need for external logic gates and ensures accurate digit representation.

I. Role of Current-Limiting Resistors:

Each segment of the 7-segment display is connected through a current-limiting resistor. These all resistors are used so they protect the LEDs from excessive current, prevent damage to the display, and ensure uniform brightness across all segments.

J. Truth Table:

Decimal Digit	BCD Input (D C B A)	Segments to be ON (a' b' c' d' e' f' g')
0	0000	0000001
1	0001	1001111
2	0010	0010010
3	0011	0000110
4	0100	1001100

5	0101	0100100
6	0110	1100000
7	0111	0001111
8	1000	0000000
9	1001	0001100
Invalid BCD	1010-1111	x x x x x x x

K. Simplified Boolean Expressions:

- $a = A'C' + DC + B' + A + C$
- $b = B' + A + DC + D'C'$
- $c = A' + B' + D + C'$
- $d = D'C + AD' + AC + A'DC'$
- $e = D'C + A'C$
- $f = B' + A'C + DC$
- $g = B' + A'D + D'C + AD'$

VI. DESIGN METHODOLOGY

A. Overall System Design Explanation:

The overall design of the 7-segment counter display is divided into three main parts: input, processing, and output. The input module uses a push button to generate clock pulses for counting. These pulses are applied to a paired cascaded 74190 BCD counter ICs, which form the processing unit of the system. The counters generate BCD outputs for the units and tens digits. After that, the common-anode 7-segment displays represent the count from 00 to 99.

B. Input Module Explanation:

The input module consists of a push button used to generate clock pulses for the counter ICs. Every press of the button increments the count by one. The push button allows controlled and observable counting, making it suitable for demonstration (and testing). In practice, the button sometimes introduce signal bouncing, so careful operation are maintained to ensure stable clock pulses. This input method replaces the manual BCD selection, and enables automatic counting behaviour.

C. Decoder Logic Operation:

The BCD outputs generated by the 74190 counter ICs are applied to corresponding 7447 decoder ICs. Each decoder processes a 4-bit BCD input and uses internal combinational logic that determines which segments of the 7-segment display should be activated. The decoder outputs are active-Low, which means segments are glowing only when a LOW signal is applied. This configuration ensures that the correct decimal digit is displayed for both the units and tens positions without the need for external logic gates..

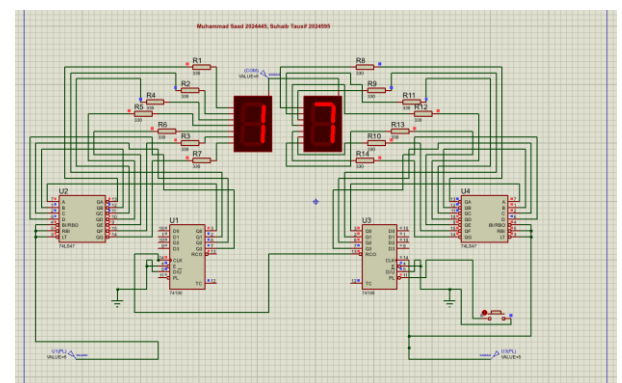
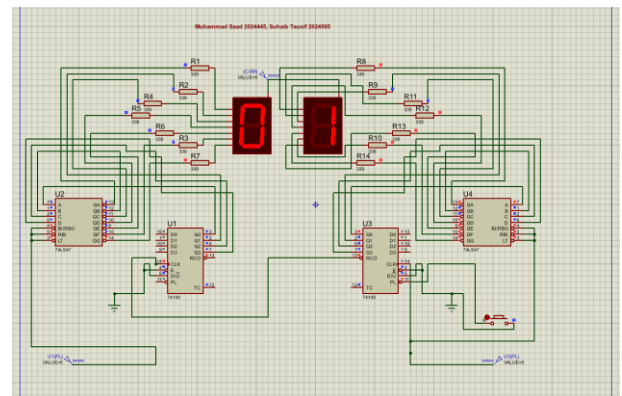
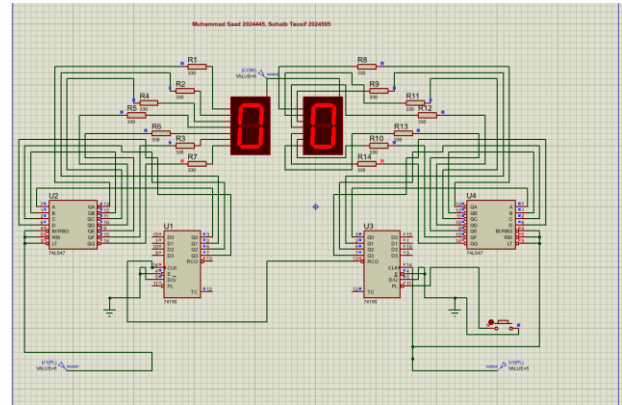
D. Output Interfacing:

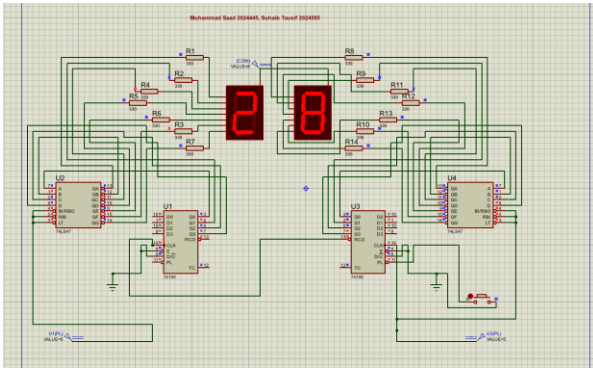
The output module consists of two common-anode 7-segment displays are connected to the 7447 decoder outputs through current-limiting resistors. These resistors protect the LED segments from excessive current. Each decoder output pin directly controls its respective segment. Using the two displays allows the

system to present a two-digit decimal number, improve readability, and extend the counting range.

E. Circuit Diagram Explanation:

The complete circuit is designed and simulated using Proteus software. The push button is connected to the clock input of the first 74190 counter. The cascading connections link the units and tens counters. The BCD outputs of each counter are connected to their respective 7447 decoder ICs, whose outputs drive the 7-segment displays through resistors. The circuit diagram clearly shows power connections, grounding, clock routing, and segment wiring.





F. Components:

The project requires standard digital electronics components to build a reliable two-digit 7-segment counter display. Each component is that is selected, based on its specific role in correct counting, decoding, and displaying of decimal numbers. Understanding the purpose of each component is important for proper assembly, testing, and troubleshooting of the circuit.

The following table summarises the components name, quantity, specifications, and purpose to provide a clear overview for project implementation.

Component Name	Quantity	Specification	Purpose
7447 IC	2	TTL BCD to 7-segment decoder	Decodes BCD outputs to display signals
74190 IC	2	TTL BCD up/down counter	Generates BCD count for units and tens
Push Button	1	Momentary switch	Provides clock pulses for counting
Resistors	14	330 Ω each	Current limiting for LED segments
7-Segment Display	2	Common anode	Displays decimal digits
Power Supply	1	+5V regulated	Provides a stable voltage to the circuit

G. 7447 IC (decoding):

It Converts 4-bit BCD output from the counter ICs into corresponding 7-segment display signals.

H. 74190 IC (counting):

It Generates BCD count values and enables cascading to implement a two-digit counter.

I. Push button (clock input):

Push button provides manual clock pulses to increment the counter in a controlled manner.

J. Resistors (current limiting):

It Protect LED segments from excessive current and ensure uniform brightness across displays.

VII. IMPLEMENTATION AND TESTING

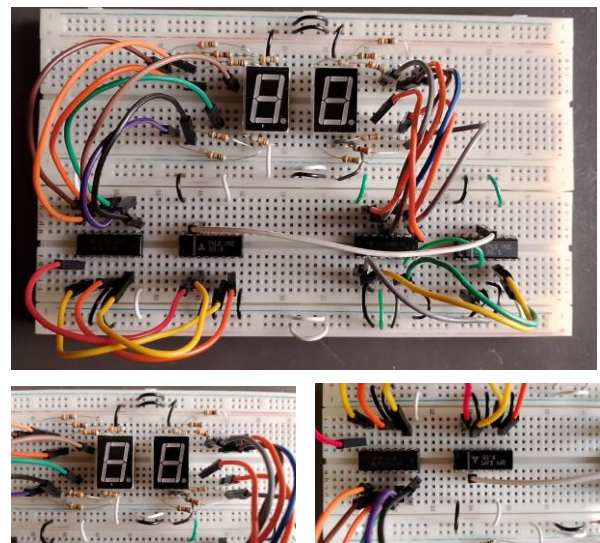
The assembly of the two-digit 7-segment counter display was carried out on a breadboard. First, both 74190 counter ICs were placed on the breadboard. The power and ground pins were connected to a regulated +5V supply. The first counter was configured as the unit counter, while the second counter was tumble to function as the tens counter. Proper carry connections were made for automatic increment of the tens digit after every ten counts.

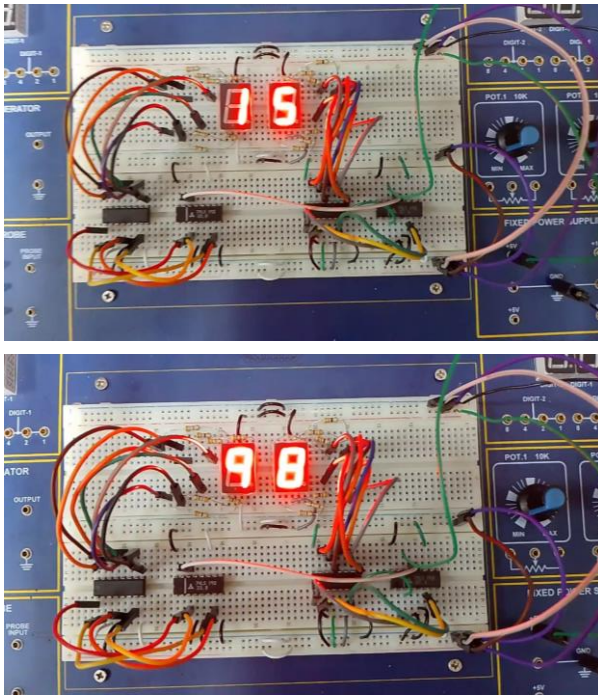
Next, the two 7447 decoder ICs were placed on the breadboard. The BCD outputs of each 74190 counter were connected to the corresponding input pins of the respective 7447 decoder and current-limiting resistors of 330 Ω were connected in series with each segment of the two common-anode 7-segment displays. The output pins of each 7447 decoder were connected to the appropriate segments of the displays. A push button was connected to the clock input of the unit counter. All connections were carefully checked for correctness, IC orientation, and secure wiring before powering the circuit.

A regulated +5V DC power supply was used for the entire circuit to ensure stable operation of the TTL ICs and displays. Proper grounding was maintained throughout the circuit to prevent floating inputs and unreliable behavior.

The testing procedure involved pressing the push button repeatedly while observing the displayed output on the two 7-segment displays. Each button press incremented the number on 7-segment displays by one. The displayed numbers were observed to progress (sequentially) from 00 up to 99. The observed output was compared with the expected counting sequence to verify the correct operation of both the counters and decoders.

At the end, the observed results matched the expected decimal count for all values from 00 to 99. Minor issue, such as inconsistent button response, was resolved by rechecking through the clock. After it, the system produced a stable and accurate two-digit count display.





VIII. RESULTS AND DISCUSSION

The output of the two-digit 7-segment counter display was systematically observed during testing. The system was verified by applying successive clock pulses through the push button, monitoring the displayed count on both the units and tens displays. The below table concludes the observed counting behaviour:

Count Value	Displayed Output
Initially	00
After 1 pulse	01
After 9 pulses	09
After 10 pulses	10
After 25 pulses	25
After 99 pulses	99

The system correctly displayed the corresponding decimal number for and each count value,. It demonstrated

that the cascaded 74190 counters and 7447 decoders are working as expected. This confirmed the correctness of the sequential counting operation, proper cascading between units and tens counters, and accurate decoding for display.

IX. CONCLUSION

The two-digit 7-segment counter display project successfully demonstrated the conversion of sequential BCD outputs into human-readable decimal numbers using two 7447 decoder ICs. The system correctly displayed numbers from 00-99 on two common-anode 7-segment displays, confirming proper operation of the counters, decoders, and display modules.

All objectives are met, including the application of combinational and sequential logic, the implementation of the TTL hardware, the display of decimal digits, and the practical verification of results through step-by-step testing.

Through this project, essential skills in digital circuit assembly, cascading counters, interfacing ICs with displays, troubleshooting the wiring issues, and ensuring stable clock input were developed. The project provided a practical understanding of BCD counting, decoding, and display interfacing, and reinforced theoretical concepts through hands-on implementation.

The project not only achieved its design goals but also explained enhancements over a single-digit system, including two-digit counting and automatic incrementing, that gives a more complete understanding of practical digital counters and display systems.

REFERENCES

- [1] M. M. Mano, Digital Design, 6th ed., Pearson, 2017. J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.
- [2] [2] F. Tocci, Digital Systems: Principles and Applications, 12th ed., Pearson, 2017.
- [3] [3] Texas Instruments, SN7447 BCD to 7-Segment Decoder/Driver Datasheet, 2019.
- [4] Texas Instruments, SN74190 BCD Up/Down Counter Datasheet, 2019.
- [5] [5] Labcenter Electronics, Proteus Design Suite 8.14 User Manual, 2020.